5 **(a)** In order to investigate the photoelectric effect, a student set up the apparatus illustrated in Fig. 5.1.

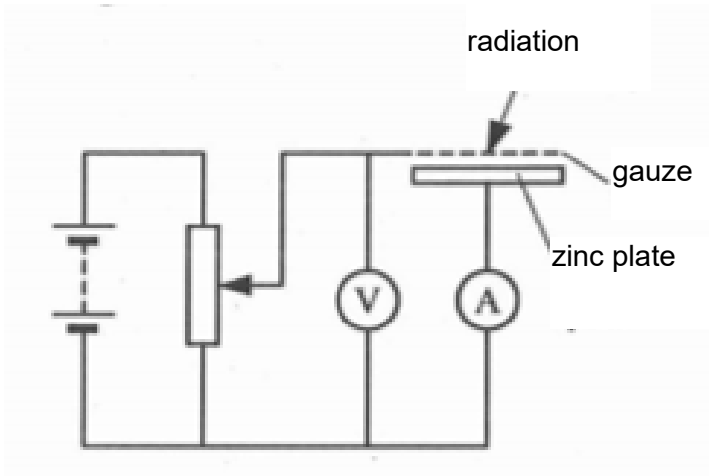

Fig. 5.1

The wavelength of the radiation incident on the zinc plate was varied. For two values of wavelength λ, the stopping voltage V_s required just to prevent electrons reaching the gauze was measured. The results are shown in Fig. 5.2.

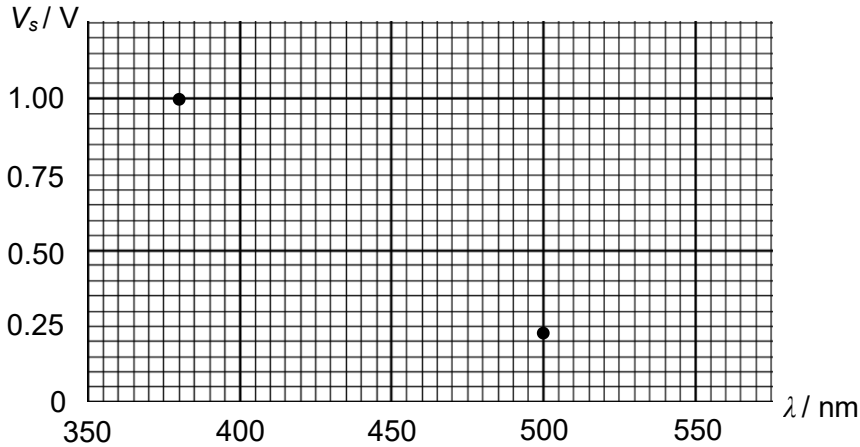

Fig. 5.2

(i) Define *work function energy*.

……

…….. [1]

(ii) Calculate the maximum speed of a photoelectron emitted from the metal surface by radiation of 380 nm.

maximum speed = …… m s^{-1} [2]

(iii) Calculate the work function energy of the metal surface.

work function energy = …… eV [2]

(iv) Suggest why it is not possible to deduce the threshold wavelength of this metal surface directly from Fig. 5.2, without further calculations.

……

…….. [1]

(b) In order to investigate the X-ray spectrum of a particular target metal, the student uses an X-ray tube where high energy electrons are fired on the target metal.

Fig 5.3 shows a sketch of the X-ray spectrum produced for the target metal. The tube voltage is 100 kV.

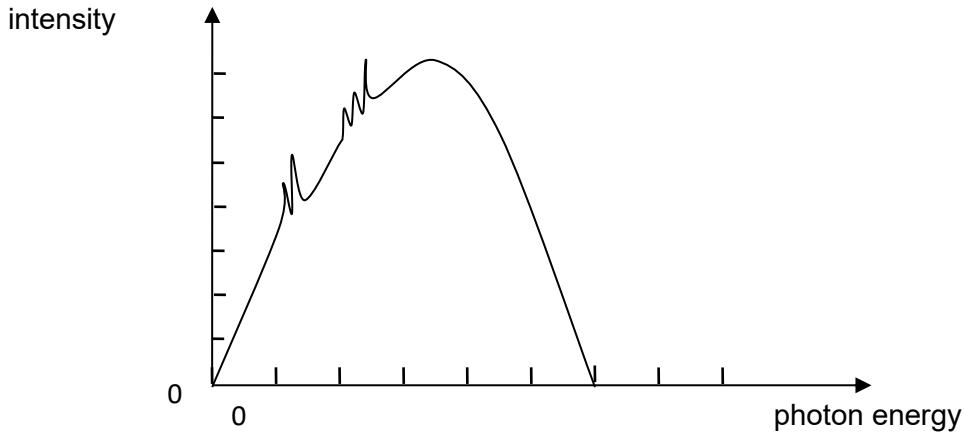

Fig. 5.3

The energy required to remove an electron from the various shells of the target metal atom is shown in Fig. 5.4.

type of shell	energy / 10^{-15} J
K	1.4
L	0.27
M	0.12

Fig. 5.4

(i) Given that the two sets of peaks in Fig. 5.3 correspond to the K and L spectrum lines, label K_α and L_β. [1]

(ii) On Fig. 5.3, sketch a spectrum for X-ray from the tube if the tube voltage is halved. [1]

(iii) Calculate the energy of the X-ray photon of longest wavelength in the K-spectrum of the target metal.

energy = …… eV [2]

(iv) Explain the formation of K_β spectrum lines.

……

……

……

…….. [2]